

Haowen (Iskar) Deng

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Education

University of Washington Master of Science in Computational Linguistics	Seattle, USA Sep. 2024 – Jun. 2026 (estimated)
Peking University Bachelor of Arts in Computational and Applied Linguistics	Beijing, China Sep. 2019 – Jul. 2024

Research Interests

My research examines how the cognitive mechanisms of language models differ from those of humans, especially in their typological biases. Using controlled synthetic languages, I investigate how factors such as data scale and grammatical cues shape their inductive biases. I aim to build mechanistic accounts of language learning linking computational modeling, linguistic typology, and cognitive science.

Core Research Projects

Differences in Typological Alignment in Language Models' Treatment of DAM Advisor: Professor Shane Steinert-Threlkeld	Seattle, USA Apr. 2025 – Dec. 2025
• Injected Differential Argument Marking (DAM) into English corpora as a controlled perturbation to test whether language models exhibit human-like inductive biases. • Examined the learnability of DAM strategies across a semantic difficulty gradient, integrating animacy, definiteness, and pronominality triggers with local vs. global rules. • Found selective alignment between LMs and human DAM universals, suggesting distinct underlying sources of these typological patterns. • Accepted to CoNLL 2026. Preprint: arXiv:2602.17653.	
Master's Thesis – Learning Nominalization: Typological Tendencies in Neural LMs Advisor: Professor Shane Steinert-Threlkeld	Seattle, USA Sep. 2025 – Present
• Generate synthetic languages with the Grammar Matrix from MRS representations extracted from parsed English corpora to study cross-linguistic nominalization strategies. • Investigate whether LMs prefer typologically plausible nominalization strategies, especially correlations between clausal complementation and action nominal constructions. • Grammar design completed; initial experiments are underway.	
Course Paper – Idiom Compositionality in Language Models	Seattle, USA Feb. 2026 – Mar. 2026
• Proposed a paraphrase-based metric to track how LM sentence representations shift between literal and idiomatic interpretations across layers. • Using TransformerLens, found that deeper layers align more with figurative meanings in idiomatic contexts, though the metric did not significantly correlate with human judgments.	
Course Project (LING 567) – Implementing Naxi Grammar with the LinGO Grammar Matrix Instructor: Professor Emily M. Bender	Seattle, USA Jan. 2025 – Mar. 2025
• Implemented an HPSG grammar for Naxi with the LinGO Grammar Matrix as part of <i>LING 567: Knowledge Engineering for NLP</i>. • Modeled key morphosyntactic phenomena in Naxi, including differential case marking, the modal verb system, and nominalization strategies.	

Other Research Projects

Multilingual Summarization of Code-Switched Social Media Content Course Project Lead	Seattle, USA Mar. 2025 – Jun. 2025
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- Developed a multilingual summarization system for code-switched Reddit posts using mT5, implementing translation-based data augmentation that improved cross-lingual generalization.

Two-Stage Prompting for Factivity Inference in Chinese (CCL 2025 Eval Task 4)	Seattle, USA Mar. 2025 – May 2025
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- Developed a two-stage prompting system combining predicate semantics and contextual reasoning with Auto-CoT, achieving 86% accuracy on the CCL 2025 factivity inference benchmark using Qwen-Max.

Undergraduate Thesis – Evaluating LLMs on Recognizing Irony in Quantitative Sentences Advisor: Professor Weidong Zhan	Beijing, China Feb. 2024 – May 2024
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- Investigated irony in sentences with quantitative categories and proposed a new irony explanatory framework.
- Built a small evaluation dataset and compared human and LLM performance on irony recognition.

Course Paper – How Large Language Models Understand Pragmatic Presupposition Advisor: Professor Wenxi Ye	Beijing, China Sep. 2023 – Feb. 2024
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- Tested large language models on presupposition types such as implicature, modality, and encyclopedic knowledge.

Term Paper – BERT-based Word Sense Disambiguation of Function Words	Beijing, China Apr. 2023 – Jun. 2023
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- Built a weakly-supervised BERT model to disambiguate senses of the Chinese adverb *cái* based on syntactic cues.

Internship Experience

China Tax Standard Certified Tax Agents Co., Ltd.	Beijing, China Sep. 2023 – Jan. 2024
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- Contributed question generation and evaluation design for a FastLLM-based tax QA system.
- Focused on diverse query formulation, server stress testing, and reliability assessment under real-world deployment conditions.

Achievements & Awards

Third-Class Scholarship of Peking University	2021–2022
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Academic Activities

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| • Participated in the CCL 2025 Evaluation Task on Factivity Inference; produced a technical report. | 2025 |
| • Attended NASSLLI 2025 (North American Summer School in Logic, Language, and Information). | 2025 |
| • Attended NeurIPS 2025 (Conference on Neural Information Processing Systems). | 2025 |
| • Selected for the “Weiming Scholars” Program at Peking University. | 2021-2023 |

Technical Proficiencies

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- **Programming:** Python, C/C++, C#
 - **Linguistic Tools:** spaCy, HPSG, LinGO Grammar Matrix, MRS
 - **Modeling & Training:** PyTorch, Hugging Face Transformers, TransformerLens
 - **Prompting & LLM Integration:** LangChain, function calling, evaluation design
 - **Linguistics:** Syntax, Typology, Semantics, Pragmatics, Phonetics
 - **Languages:** Mandarin Chinese (Native), English (Proficient)